The Evaluation of Environmental Features that Explain Energy Burden Across LA County: A Random Forest Approach

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Intro/Background

- Energy Consumption isn't just behavior
    - Income shapes consumption
    - Energy burden: % of income spent on energy
- Environmental factors
    - Greenspace and water regulate temperature
    - Extreme heat/cold drive up energy demand
- Why this study?
    - Systemic concerns, sustainability, cost, & resource management
    - Understanding urban patterns at a smaller scale

Research Question

Does the inclusion of environmental features (ie. greenspace and water body presence) improve the classification of energy burden levels of residents' across different LA County cities?

Methods: Data

- Energy burden: Low-Income Energy Affordability Data (LEAD)
- Population: Development of housing unit population distributions through the selection of cohorts assumed to have homogeneous energy use characteristics.

Variable	Operationalize	Dataset	Measure	Type
Outcome				
Energy Burden	Energy burden is the percentage of gross household income spent on energy costs.	Low-Income Energy Affordability Data	- Sum of average household annual electricity and gas expenditure divide by household income	Categorical: - Low burden < 6% - 6% ≥ High burden ≤ 10% - Severe burden ≥ 10%

Methods: Data

Variable	Operationalize	Dataset	Measure	Type
Features				
Trees	- Number of trees maintained by the LA County per city	Los Angeles County Public Works Road Maintenance Division Tree Inventory (LA county Data Catalog)	- Sum of trees in city	- Numeric
Greenspace	- Number of parks per city - The average acres of parks	Countywide Parks and Open Space (LA county Data Catalog)	- Sum of parks in cities	- Numeric
Bodies of Water	- The presence of lakes	LA County Boundary (LA county Data Catalog)	- Dichotomy of the presence of a lake	- Categorical

Method: Analytic Plan

- Used R and Rstudio
- Uneven distribution of severely burden households
  - Low = 638,000, High = 42,000, Severe = 39,000
- Stratified random sample
  - n = 20,000
- Missing data handling
  - Only complete data was considered

Method: Analytic Plan

- Applied randomForest (Breiman's original algorithm)
  - Random forest is a machine learning method that splits data into smaller groups, decision trees.
  - Using decision trees we aim to improve the classification of household energy burden.
  - Help identify patterns, predictions, and most important features that are related to energy burden

Method: Analytic Plan

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  - 3 hyperparameters
    - Number of trees
    - Number of features randomly sampled as candidates at each split was
    - Node size was varied between

Method: Analytic Plan

- Applied randomForest (Breiman's original algorithm)
  - 3 hyperparameters
    - Number of trees = 500
    - Number of features randomly sampled as candidates at each split was = 2 & 3
    - Node size was varied between = 10 & 25 (default 1)

Results

- Poor model performance
  - Out of Bag (OOB) error rate = 65%
  - Predictors had weak classification accuracy
- Confusion Matrix

	Low	High	Severe	Class error
Low	11789	814	1397	11%
High	17180	1267	1553	93%
Severe	16956	1202	1842	90%

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- Variable Importance Metrics
  - Mean decrease accuracy
    - When a feature is shuffled around using permutation how impacted are the classification?
  - Mean decrease Gini
    - How much does a feature help with the impurity of class, or unmixing of classes.

Results

Feature	Mean Decrease Accuracy	Mean Decrease Gini
Trees	20.5%	47.3%
Presences of Lake	4.6%	5.4%
Presence of Parks	16.6%	38.1%
Average Acares of Parks	30.8%	92.7%

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Discussion

- Key Findings
  - Environmental features (tree, parks, water, acreage matter BUT not strong predictors on their own
  - Energy burden requires a more complex model
  - Environmental effects (shade, temp) help but limited
- Application
  - Must go beyond environmental changes to address financial disparities
- Limitations
  - Role of water features and gaps in tree data
  - High error rate relation to predictors

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